GENEX APPLICATION

Software Engineering & Client-Side Technology Stack Specification

REACT 18 + TYPESCRIPT | OFFLINE-FIRST PWA | CLIENT-SIDE COMPUTER VISION

Document: GENEX-APP-SPEC-2026
Software Build: v2.4 (Production)
Runtime: Standalone Web App
Date: September 2026

1. Software Architecture & Client-First Paradigm

The **Genex Application** is built on a 100% *client-first, zero-backend-dependency* paradigm. All image acquisition, computer vision colorimetry, algorithmic diagnostic inference, and medical document generation execute directly within the browser runtime on the user's smartphone or PC. This guarantees sub-second responsiveness, absolute patient data privacy (HIPAA/DISHA compliant), and uninterrupted operation in low-resource clinics without internet connectivity.

Zero-Cloud Edge Processing

< 100MS

Local browser computation eliminates cloud server latency and recurring subscription API costs.

- **On-Device Pixel Analysis:** Raw camera frames are analyzed in-memory via HTML5 Canvas buffers.
- **Privacy by Design:** Zero patient images or PHI (Protected Health Information) leave the device.

Offline-First PWA Engine

100% OFFLINE

Native app installability with immediate cache-first operational readiness.

- **Service Worker Cache:** Pre-caches application bundles for offline rural OPD deployment.
- **Local Persistence:** IndexedDB / LocalStorage stores patient history and sensor calibration.

2. Frontend & Framework Technology Stack

The user interface is engineered with a modern, strictly-typed reactive framework designed for rapid clinical execution and error-free operator interaction.

Core Runtime & Build Tools

- **React 18 / 19:** Declarative component architecture managing complex diagnostic stages, interactive modals, and real-time state transitions.
- **TypeScript 5.x:** Strict static type checking across all clinical data structures, optical density math, and diagnostic payload contracts.
- **Vite 8.x:** High-performance ES module bundler delivering sub-400ms production builds and instant hot module replacement.
- **Tailwind CSS 3.4:** Utility-first styling engine enforcing responsive layout grids, custom clinical color tokens, and zero screen overflows.

UI/UX Innovations & Design System

- **Dual-Perspective Architecture:** Instant toggle between Doctor PDF Sign-Off view and Patient Plain Summary (English & Hindi).
- **Mobile-First Thumb Ergonomics:** Floating bottom navigation bar and 44px+ touch targets optimized for gloved clinic operation.
- **SVG Vector Kinetics:** High-precision interactive time-series charts rendering 0–30 min bacterial lysis vs replication curves.
- **Lucide React Icons:** Tree-shaken, ultra-lightweight clinical and diagnostic iconography.

3. In-Browser Computer Vision & Color Science

The app implements a full computer vision pipeline running directly on HTML5 Canvas without requiring heavy external computer vision servers or OpenCV binaries.

Optical Pre-Flight Quality Filters

- **Laplacian Variance Sharpness:** Computes edge gradient variance to reject motion-blurred or soft-focus photos automatically.
- **Ambient Illuminance (Lux):** Measures relative luminance to verify clinic lighting is ≥ 200 lux.
- **Specular Glare Filter:** Analyzes pixel highlight saturation to flag direct fluorescent or lamp reflections.

Scientific Colorimetric Algorithms

- **CIE L*a*b* Color Space:** Converts raw sRGB data into device-independent perceptual color coordinates.
- **CIE76 Delta-E (ΔE):** Computes Euclidean color distance against reference blanks for pyuria WBC quantification.
- **Optical Density (OD):** Logarithmic absorbance calculation: OD = -log10(I_sample / I_blank).

4. Progressive Web App (PWA) & Storage Architecture

Service Worker & PWA Manifest

- **Service Worker (sw.js):** Employs a stale-while-revalidate strategy for instant offline launches without an active network connection.
- **Web App Manifest:** Configured for standalone display mode, maskable icons, and medical theme color (#D98E3F).
- **Native Install Triggers:** Desktop and mobile install prompts for Android, iOS Safari, and Chromium browsers.

Client-Side Health Vault Storage

- **LocalStorage & IndexedDB:** Manages patient profiles, MRNs, past diagnostic sessions, and baseline calibration values.
- **Zero Data Leakage:** Full compliance with digital health privacy frameworks (DISHA, HIPAA, and GDPR).
- **Surveillance Anonymizer:** Locally strips PII before populating regional AMR heatmaps and antibiograms.

5. Clinical Documentation & Grassroots OPD Outputs

A4 Lab-Grade PDF Generation

- **jsPDF & html2canvas:** Client-side synthesis of formal, signed diagnostic certificates with verification QR codes.
- **Clinical Metadata:** Includes CLSI M100 breakpoints, MIC values, and ICMR AMRSN surveillance alerts.

OPD Thermal Slip & WhatsApp Rx

- **ESC/POS Thermal Printing:** Isolated CSS @media print layout for 58mm / 80mm portable receipt printers.
- **WhatsApp API (wa.me):** Preformatted bilingual (English & Hindi) prescription summaries sent directly to patients in 1 tap.

6. Software Technology Stack Master Reference Table

Software Layer	Technology / Library	Version / Spec	Specific Application Function
Frontend Framework	React + TypeScript	v18.3+ / 5.x	Component-driven SPA architecture, custom hooks, and strict type-safe state
Build & Bundler	Vite	v8.2	Sub-second compilation, modern ES module bundling, tree-shaking
UI Styling & Layout	Tailwind CSS	v3.4	Responsive mobile-first utility styling, custom medical color tokens, zero overflow
Iconography	Lucide React	Latest	Lightweight, scalable vector icons for clinical status, alerts, and navigation
Camera Acquisition	HTML5 MediaStreams API	W3C Standard	Direct smartphone camera stream acquisition with autofocus reticle overlay
Computer Vision Engine	HTML5 Canvas API	W3C Standard	In-browser image buffer processing, pixel sampling across 26 micro-wells
Image Quality Analysis	Custom Laplacian Algorithm	On-Device JS	Automated sharpness variance check, glare filter, and ambient lux validation
Color Science & Math	CIE L*a*b* & Delta-E	CIE76 Standard	Perceptual color distance (ΔE) and optical density (OD) calculation
Offline PWA Engine	Service Worker API	W3C PWA Spec	Pre-caches application assets; delivers 100% offline functionality in remote clinics
Local Data Storage	LocalStorage / IndexedDB	Client Storage	Stores patient health records, scan history, and white-balance calibration locally

Software Layer	Technology / Library	Version / Spec	Specific Application Function
Vector Data Graphics	Inline SVG (React)	Scalable Vector	Real-time interactive AST kinetic growth curves (0–30 min) and cartridge schematic
Certified PDF Export	jsPDF + html2canvas	Client-Side	Generates downloadable signed diagnostic reports with QR verification
OPD Thermal Printing	CSS @media print	ESC/POS 58/80mm	High-contrast monospace receipt slip formatted for clinic thermal printers
Patient Messaging	WhatsApp URI Protocol	wa.me API	Dispatches preformatted English/Hindi prescription summaries to patient phones
Standards Alignment	**CLSI M100 / ICMR AMRSN**	**Ed. 34 / 2024**	**Evidence-based antibiotic susceptibility cutoffs and clinical resistance rules**

> **Software Engineering Summary:** The Genex application is completely self-contained and operates entirely on the client side. By removing server dependencies, the platform eliminates recurring backend hosting costs, ensures 100% data confidentiality under medical privacy regulations, and provides instant, offline diagnostic execution on any modern smartphone, tablet, or PC browser.